

Week 1 Summary

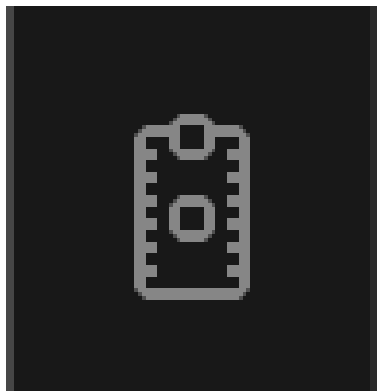
Objectives

- Receive Development Kit
- Environment Setup & Understanding Pico Environment

Activities & Implementation

- Received Raspberry Pi Pico, guided timetable, weekly report template, and internship-overview PDF
- VS Code environment setup (e.g. adding extensions)

Results



Week 2 Summary

Objectives

- Review Reference LCD Example

Activities & Implementation

- Added comments on received example codes (lcd.c & lcd.h)
- Added main function and adjusted CMakeLists.txt
 - for binary creation to make it able to print text on LCD

Results

```
const uint8_t img[16][8] = {
{ 0xFF, 0xFF, 0x00, 0x00, 0x00, 0xFF, 0xFF, 0xFF },
{ 0xFF, 0x00, 0x00, 0xFF, 0xFF, 0x00, 0x00, 0xFF },
{ 0xFF, 0x00, 0xFF, 0xFF, 0xFF, 0xFF, 0x00, 0xFF },
{ 0x00, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0x00, 0xFF },
{ 0x00, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0x00 },
{ 0x00, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0x00 },
{ 0x00, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0x00 },
{ 0x00, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0x00 },
{ 0x00, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0x00 },
{ 0x00, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0x00, 0xFF },
{ 0xFF, 0x00, 0x00, 0xFF, 0xFF, 0x00, 0x00, 0xFF },
{ 0xFF, 0xFF, 0x00, 0x00, 0x00, 0xFF, 0x00, 0x00 },
{ 0xFF, 0xFF, 0xFF, 0xFF, 0x00, 0x00, 0xFF, 0x00 },
{ 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF },
{ 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF },
{ 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF }
};    // 예시문자 큐
```

Week 3 Summary

Objectives

- Implementation of Grayscale Conversion

Activities & Implementation

- Input colour image and output converted grayscale image

Results



Week 4 Summary

Objectives

- Implementation of Binarisation
- Implementation of Bitmap-to-Dot Matrix Converter

Activities & Implementation

- BMP able to be converted into HEX (bmp to hex converter implemented)

- Converted HEX printed on LCD (test output)

Results

Week 5 Summary

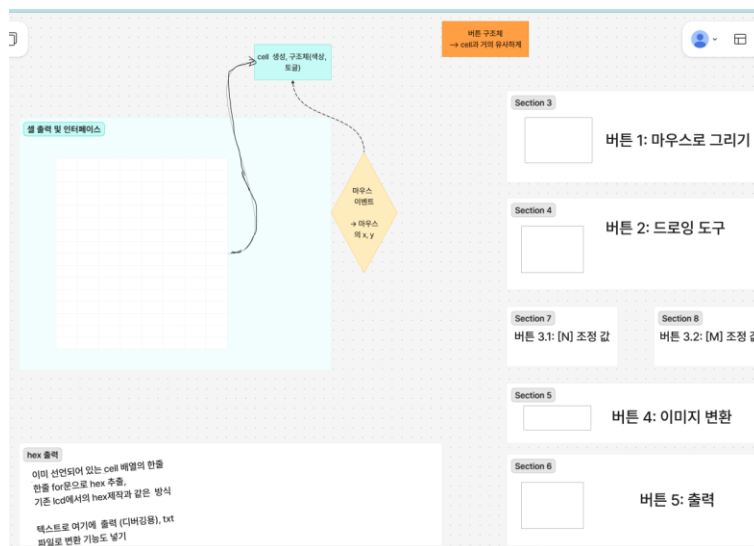
Objectives

- Environment Setup for Developing Windows GUI
- Prototyping of Windows GUI Painter

Activities & Implementation

- Windows downloaded on macOS
- MS Visual Studio set up

Results





Week 6 Summary

Objectives

- Implementation of 8x8 Dot Matrix Painter

Activities & Implementation

- Main window created
 - four buttons created with message event when clicked
 - two typeable boxes with labels created where only integers can be typed
- Added two screens
 - two main windows (wWinMain) —> one
 - layout changed: from fixed coordinates —> live layout
 - but cell sizes inconsistent (not exact square)

Results

[title here]

(button 1 name here)

(button 2 name here)

N

M

(button 4 name here)

(button 5 name here)

Info

Button 1 clicked

OK

N

(button 4 na

(button 5 na

N

M

8

8

(button 4 name here)

title here

(button 1 name here)

(button 2 name here)

N

M

(button 4 name here)

(button 5 name here)

HEX ??? ??

(button

(button

N

(button

(button

Week 7 Summary

Objectives

- Implementation of Painted Dot Matrix to Array Converter

Activities & Implementation

- Button names changed
 - Button1 name changed → “Black / White”
 - Button2 name changed → “Black Brush”
 - Button5 name changed → “OUTPUT”
- Extra button created
 - (Button3) added after Button2
- Button5 (output button) can now output paint
 - (outputs without background grids; only outputs black cells)
 - (outputs in bmp file)
 - (each outputted cell is exactly square)

Results

Black / White	
Black Brush	
N	M
[button 4 name here]	
OUTPUT	

Black / White

Black Brush

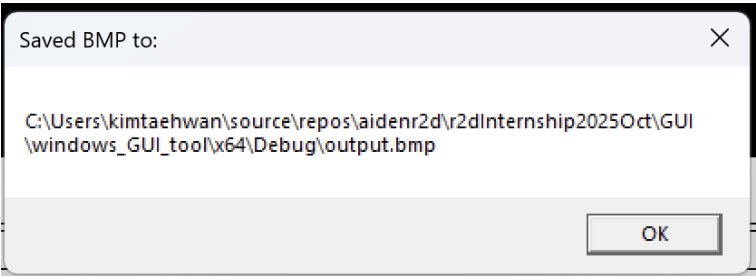
White Brush

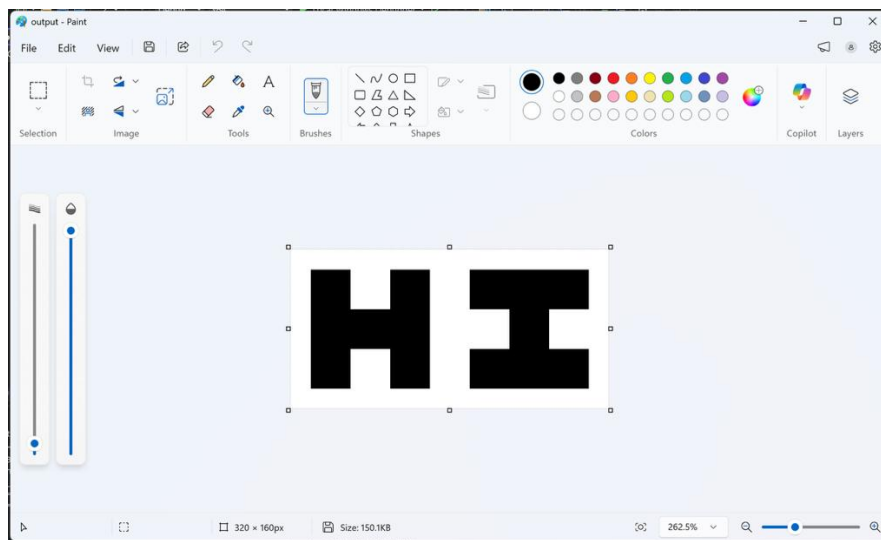
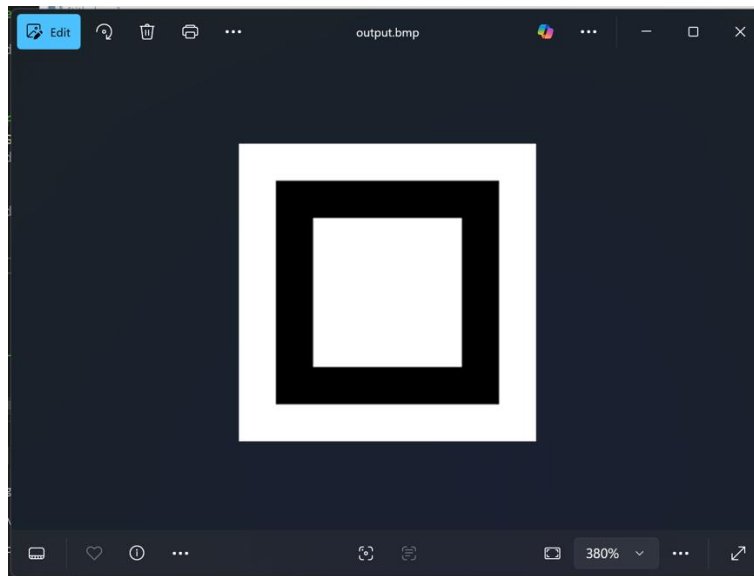
N

M

[button 4 name here]

OUTPUT





Week 8 Summary

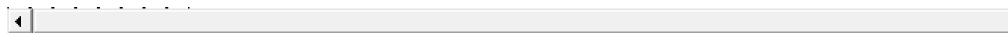
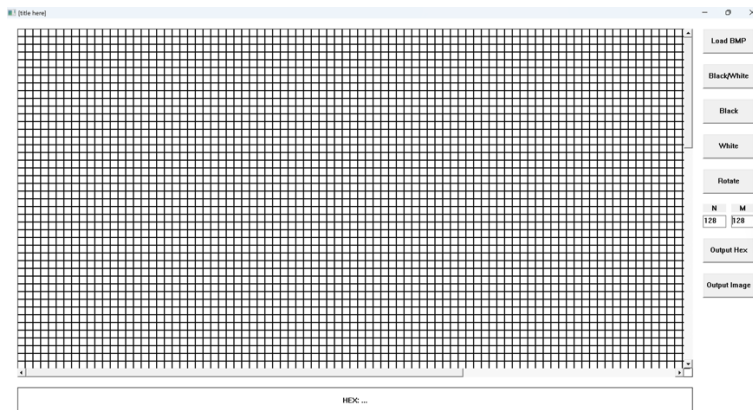
Objectives

- Implementation of $N \times M$ Scaled Dot Matrix Painter
- Wrap Up

Activities & Implementation

- Added extra button called *ButtonIn* to open a window for bmp file load (before Button1)
- Added extra button called *ButtonR* for rotating grid 90 degrees clockwise (after Button3)

- ## Results

[illegible]

